[Project Title: Insert Full Title Here]

[Student Full Name]

[High School Name], [City, State]

A Research Report Submitted for the
Regeneron Science Talent Search

[Date of Submission]

Abstract

[Insert Abstract Here — 250 words maximum. Provide a concise summary of the research question, core hypothesis or objective, computational or experimental methodology, quantitative results, and broader implications of the findings.]

1 Background of Portfolio Construction

1.1 Principles of Portfolio Construction

Modern portfolio construction involves optimizing a weight vector $w \in \mathbb{R}^n$ over n assets. To assess the optimality of a portfolio, an objective function is used. An example of a classic objective function is Markowitz's Mean-Variance framework, which minimizes a risk-adjusted utility objective:

$$\min_w(\lambda w^T \Sigma w - r^T w), \tag{1}$$

where r is an expected returns vector, λ is an exogenous risk-aversion parameter, and Σ is a covariance matrix that defines how assets move together.

Objective functions can be comprised of other forms, such as returning an L_2 norm of the difference between a calculated target weight vector $\tilde{w}$ and w.

$$f(w) = (||\tilde{w} - w||_2^2)$$

A solver would try to minimize this objective:

$$\min_w f(w).$$

This objective function will be elaborated upon in section 2.1.

Constraints encode what is permissible when calculating optimal portfolio weights. Typical constraints include individual weight and total portfolio turnovers, position limits, etc. Implementing such constraints often makes portfolio optimization non-trivial, and can present difficulties with convexity and proper execution.

Penalties occupy a middle-ground between an objective function and constraints; instead of placing a hard boundary on portfolios, they discourage going past a certain threshold continuously. In the case of (2), penalties come in the form of added terms.

Once the objective function, constraints, and penalties are defined, the choice of solving method will be contingent on the problem's structure. Objective (1) is convex if Σ is positive semi-definite and if its constraints maintain a convex feasible set. The L_2 norm objective acts similarly, placing both objectives within the reach of Disciplined Convex Programming (DCP). The convexity makes any local optimum a global minimum or maximum, and the Karush-Kuhn Tucker (KKT) conditions provide a complete characterization of optimality which we will elaborate upon later in section 2.2.

The widespread efficiency and availability of DCP solvers means convex problems can be optimized notably quickly.

1.2 Research gap related to constraints and penalties

COMPLETE LATER

2 Problem Setup

2.1 L_2 Target-Optimize Vectors

Denote the current portfolio composition as vector w^*, and denote a target composition as $\tilde{w}$, both vectors having the same size as w. Any negative values indicate shorting a stock, while positive values indicate longing a stock. $\tilde{w}$ is a signal-driven constant vector, indicating ideal target weights to consider in constructing a new portfolio. $\tilde{w}$ is normalized such that it possesses a consistent scale and is comparable across adjustments. In long-short portfolios, this typically takes the form of a gross exposure normalization:

$$\vec{1}_n^T |\tilde{w}| = \eta,$$
$$\tilde{w} = \frac{\tilde{w}}{\eta}$$

Normalizing fixes scales and units in which the objective function and subsequent constraints are expressed, and avoids degenerate solutions where magnitudes are dilated rather than truly altering relative composition of the portfolio.

Gross exposure constraints will also be placed upon w, the new portfolio being optimized. Exact gross exposure constraints that apply to w and $\tilde{w}$, ensure equal scale. This gross exposure constraint must be an equality to be effective, and is elaborated upon in section 2.6.

The primary objective function $f(w)$ we will use for the remainder of this paper will be

$$f(w) = ||\tilde{w} - w||_2^2 + (\dots), \tag{2}$$

which a solver will attempt to minimize. Additionally, $(\dots)$ indicates additional penalties that will be added to the objective in section 3. $f(w)$ is currently quadratic and convex, meaning it can be optimized through Quadratic Programming (QP) and Disciplined Convex Programming (DCP), two exceptionally efficient and quick methods many solvers utilize.

2.2 Karush-Kuhn-Tucker and Global Optimality Conditions

Because the portfolio construction problems considered henceforth are convex, their solutions are exact and can be calculated via DCP. Suppose (2) bound by affine constraints that create a convex feasible set. Consider the KKT conditions:

1. Stationarity of the Lagrangian
2. Primal feasibility
3. Dual feasibility
4. Complementary slackness

These conditions must be satisfied to prove optimality of a point satisfying constraints. An optimal point calculated by a solver will be a global optimal. Using a quadratic form calls for two conditions: a positive

semi-definite Hessian matrix, which indicates convexity, and a constraint qualification, which we will fulfill through using affine constraints.

In its current state, the objective function satisfies these conditions conditions as it contains purely quadratic terms.

2.3 Considerations when Making Constraints

Constraints can model numerous types of boundaries, including risk modeling. This includes factor exposure bounds, sector or industry neutrality, and tracking error limits. The type of risk modeling used is entirely dependent on firm-specific conditions or preferences.

Target forecasts in $\tilde{w}$ are estimated quantities, and can be noisy or anomalous. Thus, an optimizer subject to overly permissive constraints is susceptible to concreate exposure on assets with potentially inaccurate or aggressive forecasts. Turnover caps and position limits act as a form of regularization against over-adjusting to noisy estimation.

Long-short portfolios are rebalanced frequently, often incurring transaction costs. Turnover budgets and per-asset turnovers are imposed so a portfolio is discouraged from extending to marginal returns for greater transaction costs.

While constraints have utility, each imposition reduces the feasible region of potential solutions. An overly aggressive set of boundaries can force the optimizer to a suboptimal corner of a feasible set that reflects very little of the objective. Therefore we will set constraints such that they are reflective of practical and risk limits rather than as the determining factor in calculating an optimal point. In other words, we do not wish for constraints to "drive" the solution; the solver should be driven in part by $\tilde{w}$.

Regardless of a constraint's function, we will show a preference for preserving convexity of the overall feasible region. Non-convexity in a feasible region makes it far more difficult for a solver to reliably calculate an optimal solution within a brief time frame, often trapping the algorithm in local minima. Linear equalities and inequalities, as well as second-order cone constraints define half-spaces, hyperplanes, and second-order cones respectively. These all define strictly convex sets. For constraints that are non-smooth but inherently convex (such as absolute values or piecewise functions), we utilize exact epigraph reformulations with auxiliary variables. For constraints that are truly non-convex, we employ convex relaxations to approximate the feasible space while maintaining computational tractability.

2.4 Introducing all constraints

Subjecting w to minimum and maximum individual asset boundaries can be accomplished with strictly linear constraints. Constraints are defined by vectors $\underline{w}$ and $\overline{w}$, respectively.

$$w \leq \overline{w},$$
$$\underline{w} \leq w.$$

Similarly, limiting each individual asset's turnover to values denoted by vector t can be accomplished through more linear inequalities.

$$w - w^* \leq t,$$
$$-(w - w^*) \leq t.$$

Net exposure, denoted by ρ, is equal to total long exposure minues total negative exposure. Essentially, it equals the sum of all values in w. We will constrain net exposure to a specific value through a linear equality.

$$\sum_{i=1}^{n} w_i = \vec{1}_n^T w = \rho$$

There are several ways to impose constraints on attribute exposure. One of the ways is to express n assets' exposure in m attributes utilizing an $m \times n$ matrix $\mathbf{S}$. For example, an potential $\mathbf{S}$ for a 3 asset matrix in an energy and financial attribute is shown in table (1).

Table 1: Example Attribute Matrix.

	Asset A	**Asset B**	**Asset C**
Financial	0.6	0.3	0.5
Energy	0.4	0.7	0.5

A matrix $\mathbf{S}$ can be used to balance portfolios to lower and upper attribute weight vectors $\underline{s}$ and $\overline{s}$, each bound vector containing m elements.

$$\underline{s} \leq \mathbf{S} \leq \overline{s} \tag{3}$$

Total turnover bounds aggregate change in position across all assets using an L_1 constraint on the difference between w and w^*, bounded by total turnover τ.

$$||w^* - w||_1 \leq \tau \tag{4}$$

This is meaningfully different from the individual turnover constraint, as it balances a total budget, instead of individual change. In practice, individual turnover is often set to be less restrictive (such that many assets in w are not equal to their maximum turnover) than total turnover, which is often pushed to τ. If individual turnover constraints are too restrictive, then a maximum turnover is essentially implied. However, it often leads to less diverse portfolio compositions, as variation between assets is diminished.

The absolute value summation constraint $\sum_{i=1}^{n} |w_i - w_i^*| \leq \tau$ is not natively solvable in many solvers. Although the constraint is convex, it is not affine, and is not differentiable at $w_i = w_i^*$. To alleviate this, slack variables are created to reformulate the constraint. This operation is explain in detail in 2.5.

Gross exposure, denoted by η, is another value commonly utilized in portfolio optimization. It is calculated explicitly through

$$||w||_1 = \eta. \tag{5}$$

(5) is typically an equality constraint, as firms often wish to keep a specific amount of capital in a portolio.

However, creating this equality constraint explicitly results in non-convexity (to be proved later). This is different from (4). This constraint is simply unviable in its current form, while (4) is convex and can be reformulated in different expressions. Thus, we will need to find a way to reformulate or avoid this non-convexity. The convexity of (5) is addressed in section 2.6.

It's important to note that in long-only or short-only portfolios, non-convexity is not an issue as (5) can be expressed without absolute values (since all w_i have the same signs). However, in long-short portfolios, an L_1 norm is required.

2.5 Elaborating on Total Turnover

Numerous solvers do not natively accept absolute value constraint expressions, making it not directly compatible with (4). A slack variable operation is created to circumvent this. A buffer vector $y \in \mathbb{R}^n$ with the same size as w is subjected to the following constraint:

$$-(w^* - w) \leq y \leq (w^* - w)$$

y is only every bounded by $|w - w^*|$, and is not forced to be equal to turnover. What forces equality in practice is an aggregate cap:

$$\sum_i y_i \leq \tau.$$

Since increasingly y_i will only make this condition more difficult to satisfy and doesn't affect the objective value, y_i will be forced downards toward the tightest feasible value $y_i = |w_i^* - w_i|$. In doing so, (4) is replaced with strictly linear expressions. At the optimal value, then,

$$\sum_i y_i = ||w_i^* - w_i||_1 \leq \tau.$$

These reformulated constraints are strictly linear, and are thus directly compatible with many solvers.

2.6 Elaborating on Gross Exposure

Let us begin by proving that (5) is a non-convex constraint using the definition of convexity.

A set $\mathscr{C} \subset \mathbb{R}^n$ is convex if and only if for any two points a and b in $\mathscr{C}$ and every $\theta \in [0,1]$,

$$\theta a + b(1-\theta) \in \mathscr{C}. \tag{6}$$

We will prove that for fixed $\eta > 0$, the set $\mathscr{S} = \{w \in \mathbb{R}^n : ||w||_1 = \eta\}$ is not convex.

This argument does not restrict itself to properties of the L_1 norm. Rather, it uses two axioms every norm satisfies:

1. Absolute Homogeneity: $||\alpha w|| = |\alpha|||w||$ for any scalar α.
2. Positive definiteness: $||w|| = 0 \Leftrightarrow w = 0$.

Take any $w \in \mathscr{S}$. By homogeneity with $\alpha = -1$,

$$||-w||_1 = |-1| \cdot ||w||_1 = ||w||_1 = \eta,$$

so $-w \in \mathscr{S}$ as well. Now take $\theta = \frac{1}{2}$ and form (6) of w and $-w$:

$$\theta w + (1-\theta)(-w) = \frac{1}{2}w - \frac{1}{2}w = 0.$$

By positive definiteness, $||0||_1 = 0$. Since $\eta > 0$ by assumption, $0 \neq \eta$, so

$$\theta w + (1-\theta)(-w) = 0 \notin \mathscr{S}.$$

We've exhibited $w, -w \in \mathscr{S}$ and $\theta = \frac{1}{2} \in [0,1]$ with $\theta w + (1-\theta)(-w) \notin \mathscr{S}$. By (6), the set $\mathscr{S}$ is not convex.

Our circumvention to this non-convexity is forcing the sign of w_i to match the sign of $\tilde{w}$ for all $1 \leq i \leq n$. Suppose $\mathscr{W}^+$ denotes the sub-vector of w where $\tilde{w}_i > 0$ and $\eta^+ > 0$ is the desired long exposure, with $\mathscr{W}^-$ and $\eta^- < 0$ being their negative counterparts. $\mathscr{W}^0$ denote the sub-vector of w where $\tilde{w}_i = 0$. We then introduce the reformulated gross exposure constraints:

$$\begin{aligned} \vec{1}_n^T \mathscr{W}^+ &= \eta^+, \\ \vec{1}_n^T \mathscr{W}^- &= \eta^-, \\ \mathscr{W}^0 &= 0, \\ \eta^+ - \eta^- &= \eta. \end{aligned}$$

These constraints avoid the non-convexity presented by (5) through only utilizing linear constraints. Furthermore, it also presents certain practicalities. ADD HERE?

3 Additional gating

3.1 Value in Penalties

Hard constraints that bound turnover and other portfolio characteristics have certain downsides. Violating a hard cosntraint by a negligible amount and by a comparatively huge amount have the exact same effect, meaning rounding errors, tiny violations, and genuine infeasibility are treated the same. Furthermore, if for some i, $|w_i^* - \tilde{w}_i|$ is greater than t_i, there is a sizable chance that the stock should be considerably skewed in the direction of $\tilde{w}_i$. In those cases, it is desirable to permit certain violations of hard constraints. To do this, we introduce soft constraints.

Soft constraints permit violations of hard constraints while simultaneously discouraging large violations through incurring a penalty on the objective function. We will consider 3 distinct types of soft constraints: an alpha penalty, an Elastic Net, and a Berhu penalty.

3.2 Alpha gating for total upper turnover

Instead of establishing a hard upper bound τ on total turnover, with $||w_i - w_i^*||_1 \leq \tau$, we relax it to

$$||w_i - w_i^*||_1 \leq \tau + z,$$

and change the objective function to

$$||w - \tilde{w}||_2^2 + \alpha z^2,$$

where z is a single-dimension vector and α is an adjustable exogenous parameter.

This implementation is very intuitive. $||w_i - w_i^*||_1$ is allowed to exceed τ, but the objective function experiences a penalty to do so. If turnover is less than τ, no penalty is added. The larger the of α, the more the final turnover is discouraged from exceeding τ, and vice versa. With the addition of this "alpha gate", the total turnover constraint is no longer a rigid hard bound, allowing more adaptability.

3.3 Elastic Net

The change from w^* to w incurs transaction costs. Naturally, this requires balancing tracking error and turnover so as to avoid paying fixed costs on insignificant transactions. Using a concrete turnover bound that doesn't discourage/encourage different magnitudes of turnover thus has downsides. To alleviate this, we introduce the Elastic Net.

We wish to induce a degree of sparsity in a portfolio's turnovers, meaning making fewer, concentrated trades rather than making several tiny trades that are will not have a large affect on portfolio output, but incur fixed costs. The Elastic Net is a penalty on the objective that utilizes both an L_1 and L_2 norm, allowing a firm to adjust the degree of sparsity in portfolio turnover:

$$E(w) = \beta||w - w^*||_1 + \gamma||w - w^*||_2^2, \tag{7}$$

where β and γ are adjustable exogenous parameters. The L_1 penalty promotes sparsity. It is a mechanism that drives small weight turnovers to 0, reducing the number of assets to trade. The L_1 multipler β governs the degree of sparsity on the portfolio. A larger β reflects a higher marginal cost of initiating a new trade. This shrinks the amount of sensible trades, forcing the optimizer to only execute trades where expected returns outweigh a fixed transaction cost.

The L_2 penalty promotes smoothness. it prevents a single asset's weight from changing drastically. Hence, turnover among correlated assets will be distributed more evenly than with solve the absolute value penalty. For example, the L_2 penalty resulting from two turnovers 0.2 and 0.2 is smaller than the penalty resulting from 0.1 and 0.3, even though the total turnover is the same. Thus, the L_2 turnover results in spreading turnover more evenly across assets. The L_2 multipler γ governs concentration of turnover. A higher γ penalizes larger magnitudes of concentrated shifts in individual assets. Thus, the penalty favors spreading total turnover more evenly across several assets.

The ratio $\frac{\beta}{\gamma}$ is a critical dial in determining the behavior of the penalty. A high ratio discourages trading many stocks, and a low ratio prioritizes smooth rebalancing.

(7) and the alpha gating quadratic term can be added to the base L_2 objective function (2):

$$f(w) = ||\tilde{w} - w||_2^2 + \beta||w - w^*||_1 + \gamma||w - w^*||_2^2 + \alpha z^2$$

Each term in the objective is either an L_1 norm, L_2 norm, or a quadratic function. If all scalars α, β, and γ are positive (which they should be, otherwise penalties will not work as intended), the entire objective function will be strictly convex. This ensures that any optimal solution is a global optimum. Note that an L_1 norm in the objective function is permissible, while an L_1 norm equality constraint is not convex. ADD OTHER FIELDS THAT UTILIZE ELASTIC FIELDS

3.4 Berhu's (Reverse Huber's) Loss

Elastic Net penalties subject portfolio weights to both L_1 and L_2 penalties, regardless of the magnitude of an asset's turnover. We can introduce a piecewise function that contains separate L_1 and L_2 penalties, then set a specific threshold κ where one or the other is used. To do this, we introduce a Berhu's (Reverse Huber) Loss.

Denote $u_i = |w_i - w_i^*|$. The Huber's Loss is then

$$h(w_i) = \begin{cases} \frac{(u_i)^2 + \kappa^2}{2\kappa} & \text{if } u_i \leq \kappa, \\ u_i & \text{if } u_i \geq \kappa. \end{cases} \tag{8}$$

This function is commonly used in regression. It applies a quadratic penalty for small deviations, and a linear penalty for large deviations. This mitigates the influence of extreme outliers in regression while maintaining differentiability throughout.

Berhu's Loss, also known as Reverse Huber's Loss, inverts the order of the quadratic and linear functions, and is added to the objective function; for a turnover u_i and a theshold scalar $k > 0$, the Berhu Loss is defined as

$$b(w_i) = \begin{cases} u_i & \text{if } u_i \leq \kappa, \\ \frac{u_i^2 + \kappa^2}{2\kappa} & \text{if } u_i \geq \kappa. \end{cases} \tag{9}$$

This functions similarly to an Elastic Net, but only applies either an L_1 norm or L_2 norm at a time. For minor weight adjustments where $u_i \leq \kappa$, the penalty is linear, promoting sparsity. For significant weight adjustments where $u_i \geq \kappa$, the penalty becomes quadratic and prevents concentrated trades.

This function is convex and continuous everywhere. That said, its piecewise nature makes it difficult to implement directly into a solver, as many solvers do not accept piecewise functions natively. To resolve this, we'll create an epigraph formulation with auxiliary variables.

Let s_i and t_i be two non-negative auxiliary variables, subject to the following constraints:

$$\begin{aligned} s_i + t_i &\geq u_i \\ 0 \leq s_i &\leq \kappa \\ 0 &\leq t_i \end{aligned}$$

The Berhu Loss then becomes

$$b(w_i) = \sum(s_i + t_i + \frac{t_i^2}{2\kappa}), \tag{10}$$

We can prove equivalence with basic casework. In the first case, $u_i \leq \kappa$. To minimize the penalty, the solver will set $t_i = 0$, since the quadratic penalty term $\frac{t_i^2}{2\kappa}$ grows more rapidly than the linear term s_i, being more "expensive" in the objective function. The constraint $s_i + t_i \geq u_i$ forces $s_i = u_i$, making the penalty exactly u_i, matching the linear branch of (9).

In case two, $u_i > \kappa$. The solver will maximize the linear variable $s_i = \kappa$, resulting in t_i being non-zero:

$$t_i = u_i - \kappa,$$

while the cost becomes

$$\begin{aligned} &\kappa + (u_i - \kappa) + \frac{(u_i - \kappa)^2}{2\kappa} \\ =&u_i + \frac{u_i^2 - 2u_i\kappa + \kappa^2}{2\kappa} \\ =&\frac{u_i^2 + \kappa^2}{2\kappa}. \end{aligned}$$

This matches the value of the quadratic branch of (9). Thus, we've reformulated the piece-wise Berhu function (9) into only an epigraph with auxiliary variables.

(10) can then be added with a scalar to the objective function along with the alpha gating:

$$||w - \tilde{w}||_2^2 + \alpha z^2 + \lambda \sum(s_i + t_i + \frac{t_i^2}{2\kappa}). \tag{11}$$

Where λ is an adjustable exogenous scale coefficient. All terms in the objective function are either quadratic or linear, so the function is convex.

Once again, it is important to consider the impact of changes in the parameters. Threshold κ is the inflection point where the penalty transitions from linear to quadratic. A small κ means the optimizer switches to quadratic penalties at very small turnover values, resulting in a more dense portfolio with fewer turnovers pushed to exactly 0.

A large κ means the penalty remains linear for a wider range of turnovers, promoting sparsity in the same way the Elastic Net acted.

Scale λ is a global multiplier for the Berhu penalty. It controls how large of an impact the penalty has on the objective. A higher λ makes the optimizer prioritize responding to transaction costs more than perfectly matching $\tilde{w}$.

As stated before, the difference between (7) and (10) is that (10) imposes only one penalty per asset, while (7) imposes both simultaneously. So in an Elastic Net, the "effective" L_1 penalty is always slightly increased, even when u_i is almost negligible. For the Berhu's Loss, one can set a concrete threshold κ past which they wish to start imposing quadratic penalties.

ADD ADDITIONAL USAGES OF BERHU/HUBER?

4 Numerical Experiments

4.1 contents tbd

5 Huber's Loss Objective Function

The priorly stated Huber's Loss function (8) can be used as a completely different objective function, not just a penalty.

For a given deviation vector $|w - \tilde{w}| = v$, amd a threshold parameter κ, the piecewise Huber's Loss function is defined as

$$h(v) = \begin{cases} \frac{(v)^2+\kappa^2}{2\kappa} & \text{if } v \leq \kappa, \\ v & \text{if } v \geq \kappa. \end{cases}$$

Similar to (10), this can be reformulated to a single equation using auxiliary variables s_i and t_i:

$$h(v) = \lambda(\frac{s_i^2}{2\kappa} + t_i + \frac{\kappa}{2}) + (\dots), \tag{12}$$

where

$$\begin{aligned} s_i + t_i &\geq v_i, \\ 0 \leq s_i &\leq \kappa, \\ t_i &\geq 0. \end{aligned}$$

Perhaps unsurprisingly, this is the mirror opposite of our previously derived reformulation for the Berhu's Loss. (12) can then be made the entire objective function, along with other penalties such as alpha gating, Elastic Net, or Berhu's Loss. In portfolio optimization, minimizing the squared tracking error (deviation from $\tilde{w}$) or optimizing a covariance matrix σ can make the optimizer sensitive to outliers. If some $\tilde{w}$ predicts an extreme, inaccurate weight for a single stock, the optimizer can appropriate a comparatively massive amount of exposure. A Huber's Loss base objective function applies an L_2 norm for small $|w - \tilde{w}|$, and an $|L_1|$ norm for larger turnovers, preventing anomalous $\tilde{w}$ values from hijacking total exposure allocation. The value of λ dictates how much the solver will discourage responses to large outliers in $\tilde{w}$.

6 Conclusion

6.1 Parallels with Machine Learning

6.2 Next steps in research

References

[1] Steven Diamond and Stephen Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

[2] Paul J. Goulart and Yuwen Chen. Clarabel: An interior-point solver for conic programs with quadratic objectives, 2024.

[3] Art B. Owen. A robust hybrid of Lasso and Ridge regression. *Contemporary Mathematics*, 443:59–72, 2007.

[4] Roland Schwan, Yuning Jiang, Daniel Kuhn, and Colin N. Jones. PIQP: A proximal interior-point quadratic programming solver. In *2023 62nd IEEE Conference on Decision and Control (CDC)*, pages 1088–1093, 2023.